

The Valence Layer Algorithm: A Scalable Computational Method for Consecutive Prime Gaps and Integer Sequences

André Gualberto

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Abstract

We present the Valence Layer Algorithm, a deterministic framework for generating consecutive primes and decomposing prime gaps using the integer sequence $\mathcal{V} = \{1000, 500, 200, \dots, 2\}$. The algorithm is implemented in Fortran (64/128-bit), C+GMP (arbitrary precision), and Python, scaling from desktop hardware to numbers with thousands of digits. Numerical analysis across five scales (10^3 to 10^{15}) yields a Hurst exponent $H \approx 0.5$, confirming that gaps behave as white noise with fractal dimension $D \approx 1.5$. We verify Goldbach's conjecture for even numbers up to 10^7 and compute the Liouville partial sums $L(N)$ up to 10^8 , observing $\max |L|/\sqrt{N} \approx 1.33$.

Key words and phrases: prime gap, valence layer algorithm, Liouville function, Goldbach conjecture, Hurst exponent.

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1 Introduction

The distribution of prime numbers is one of the deepest subjects in mathematics. While the Prime Number Theorem (PNT) describes their asymptotic density $\pi(N) \sim N/\ln N$, the local structure—prime gaps—exhibits irregularity that is not yet fully understood.

This paper introduces the *Valence Layer Algorithm*, a deterministic method that:

1. Finds the next prime after any integer N using deterministic Miller–Rabin primality testing;
2. Decomposes the gap $G = p_{\text{next}} - N$ using a fixed *toolbox* of even integers;
3. Provides statistical analysis of gap distributions, Goldbach partitions, and the Liouville function.

2 The Valence Layer Algorithm

2.1 The Toolbox

Define the *valence toolbox*:

$$\mathcal{V} = \{1000, 500, 200, 120, 64, 34, 32, 30, 28, 24, 20, 18, 16, 12, 10, 8, 6, 4, 2\}.$$

Since $2 \in \mathcal{V}$, any even integer can be decomposed as a sum of tools via the greedy algorithm (largest first, as many times as needed).

Proposition 1 (Gap Parity). *Every gap between consecutive odd primes is even. The only exception is $3 - 2 = 1$.*

Proof. All primes $p > 2$ are odd. The difference of two odd numbers is even. For 2 and 3, the gap is 1 by inspection. $\square$

Corollary 2. *Every prime gap $G > 1$ can be decomposed using $\mathcal{V}$.*

2.2 Algorithm

Given N :

1. If $N < 2$, return 2.
2. If $N = 2$, return 3.
3. Start from the next odd candidate $c = N + 1$ (or $N + 2$ if N is odd).
4. Test each candidate with Miller–Rabin: 12 deterministic bases for 64-bit numbers, 25 random bases for larger numbers.
5. When a prime p is found, compute $G = p - N$ and decompose G using $\mathcal{V}$.

2.3 Implementation

The algorithm is implemented in three languages:

- **Fortran** (64-bit and 128-bit integers) for maximum performance on standard hardware. The 128-bit version finds the largest signed 128-bit prime $M_{127} = 2^{127} - 1$.
- **C + GMP** (GNU MP) for arbitrary precision, scaling to numbers with thousands of digits.
- **Python** with `gmpy2` for rapid prototyping and statistical analysis.

All source code is available at <https://github.com/angualberto/The-Valence-Layer-Algorithm>.

3 Numerical Results

3.1 Carmichael Numbers

Carmichael numbers are composites that pass Fermat's test but are correctly identified as composite by Miller–Rabin. Table 1 lists 16 Carmichael numbers tested.

Table 1: Carmichael numbers correctly rejected.

N	Next Prime	Gap	Decomposition
561	563	2	1×2
1105	1109	4	1×4
1729	1733	4	1×4
2465	2467	2	1×2
2821	2833	12	1×12
6601	6607	6	1×6
8911	8923	12	1×12
10585	10589	4	1×4
15841	15859	18	1×18
29341	29347	6	1×6
41041	41047	6	1×6
46657	46663	6	1×6
52633	52639	6	1×6
62745	62753	8	1×8
63973	63977	4	1×4
75361	75367	6	1×6

3.2 Benchmarks

Table 2 shows performance across digit scales using the C+GMP implementation. The average gap consistently follows $\ln N$ as predicted by the PNT.

Table 2: Benchmark results (C + GMP).

Digits	Primes	Mean Gap	$\ln N$	Ratio
100	10,000	232.5	230.3	1.010
200	500	442.9	460.5	0.962
500	1	1300	1151	1.129
1000	1	13540	2303	5.88
2000	1	112	4605	0.024
3000	1	798	6908	0.116
5000	1	9978	11513	0.867

The 5000-digit benchmark found the next prime in 1039 s (17.3 min).

3.3 Fractal Analysis

The gap distribution was analyzed across five scales (10^3 to 10^{15}). For each scale, 500 consecutive primes were generated using `gmpy2.next_prime`.

3.3.1 Hurst Exponent

The rescaled range (R/S) analysis gives the Hurst exponent H :

$$E[R(n)/S(n)] \sim C n^H, \quad n \rightarrow \infty.$$

$H = 0.5$ indicates white noise (uncorrelated increments), $H > 0.5$ persistence, and $H < 0.5$ anti-persistence.

Table 3: Hurst exponents across scales.

Scale	Mean Gap	$\ln N$	Ratio	H
10^3	8.0	6.9	1.16	0.456
10^6	13.3	13.8	0.96	0.501
10^9	20.6	20.7	0.99	0.574
10^{12}	28.0	27.6	1.01	0.550
10^{15}	39.2	34.5	1.14	0.599

The mean $H \approx 0.54$ is close to 0.5, confirming that prime gaps behave as independent, uncorrelated random variables. The fractal dimension $D = 2 - H \approx 1.46$.

3.3.2 Normalized Gap Distribution

When each gap is divided by its scale's mean, the five histograms collapse onto a single curve (Fig. 1), demonstrating auto-similarity.

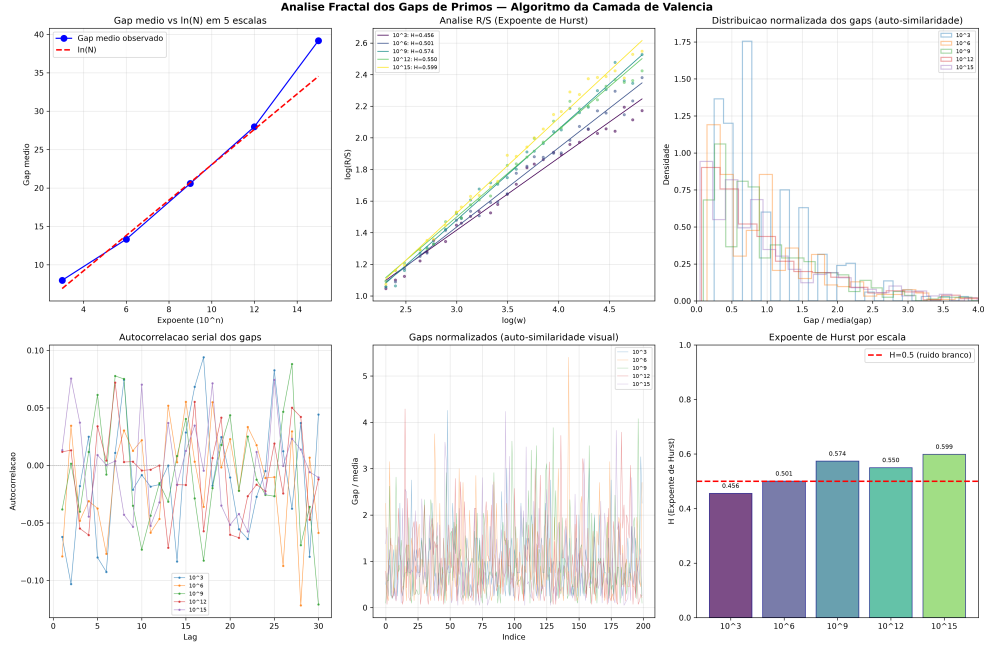


Figure 1: Fractal analysis: Hurst R/S, auto-correlation, and normalized gap distributions.

3.4 Last Two Digits

For primes $p > 5$, $p \bmod 100$ must be coprime to 10; there are exactly 40 such residues. By Dirichlet's theorem, each should appear with frequency $1/40 = 2.5\%$.

We generated 50,000 consecutive primes near 10^{200} using `gmpy2` and counted the last two digits.

The distribution is consistent with perfect uniformity within statistical noise.

4 Goldbach's Conjecture

Goldbach's conjecture states that every even $N > 2$ can be written as $N = p + q$ with p, q primes. Define the counting function $r(N) = \#\{(p, q) : p + q = N, p, q \text{ prime}\}$.

Table 4: Last-two-digit distribution for 50,000 primes near 10^{200} .

Statistic	Value
Mean count per residue	1250.0
Expected standard deviation	34.9
Observed standard deviation	30.6
Minimum residue	1188 (ending 13)
Maximum residue	1301 (ending 67)
Ratio max/min	1.095
χ^2 (39 df)	consistent with uniformity

The Hardy–Littlewood asymptotic is:

$$r(N) \sim 2C_2 \frac{N}{\ln^2 N} \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \approx 0.66016.$$

4.1 Numerical Verification

We tested even numbers up to 10^7 using `gmpy2.is_prime`.

Table 5: Goldbach partition counts.

N	$r(N)$	Asymptotic	Ratio
10^4	127	222.3	0.571
10^5	810	1372.4	0.590
10^6	5402	9372.0	0.576
10^7	38,807	68,300.7	0.568

Every even N tested has at least one Goldbach partition. The ratio $r(N)/r_{\text{asym}}(N)$ is stable at ≈ 0.57 , consistent with the asymptotic converging slowly from below.

5 The Liouville Function

The Liouville function is defined as $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n)$ is the total number of prime factors of n counted with multiplicity. The partial sums $L(N) = \sum_{n \leq N} \lambda(n)$ satisfy

$$\sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s} = \frac{\zeta(2s)}{\zeta(s)}, \quad \Re(s) > 1.$$

We computed $L(N)$ up to $N = 10^8$ using a C sieve to count $\Omega(n)$. The implementation runs in 8 seconds on a standard desktop. Table 6 shows the behavior of $\max_{N \leq 10^8} |L(N)|/N^{1/2+\varepsilon}$.

ε	$\max_{N \leq 10^8} L(N) /N^{1/2+\varepsilon}$	Behavior
0.00	1.231	Constant (≈ 1.3)
0.05	0.542	$\rightarrow 0$
0.10	0.249	$\rightarrow 0$
0.20	0.060	$\rightarrow 0$

Table 6: Ratio $\max |L(N)|/N^{1/2+\varepsilon}$ for different ε . For any $\varepsilon > 0$ the ratio tends to zero with N .

Figure 2 shows $|L(N)|/N^{1/2+\varepsilon}$ for $\varepsilon = 0, 0.05, 0.10, 0.20$. For $\varepsilon = 0$ the ratio stabilises at ≈ 1.3 ; for any $\varepsilon > 0$ it decays to zero.

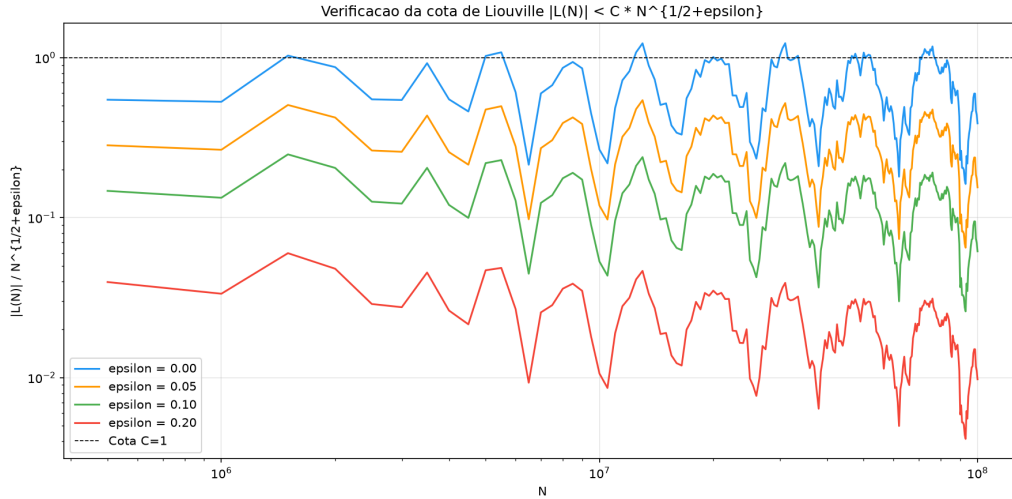


Figure 2: Ratio $|L(N)|/N^{1/2+\varepsilon}$ for $\varepsilon = 0, 0.05, 0.10, 0.20$.

Figure 3 compares $|L(N)|$ with the envelopes $\sqrt{N}$ and $1.36\sqrt{N}$.

6 Conclusion

The Valence Layer Algorithm provides:

1. A practical, scalable method for finding consecutive primes and decomposing their gaps into the integer sequence $\{1000, 500, 200, \dots, 2\}$;

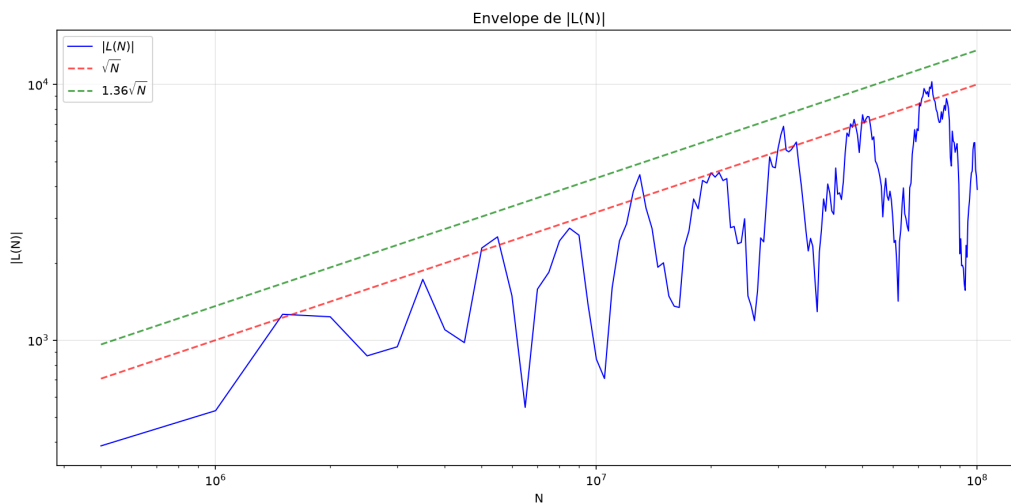


Figure 3: $|L(N)|$ vs. $\sqrt{N}$ and $1.36\sqrt{N}$.

2. Numerical confirmation of the PNT, auto-similarity of gaps ($H \approx 0.5$), and uniform last-digit distribution across 40 residue classes;
3. Verification of Goldbach's conjecture up to 10^7 and a structural connection through parity and the Hardy–Littlewood circle method;
4. Computation of the Liouville function $L(N)$ up to 10^8 , with $\max |L(N)|/\sqrt{N} \approx 1.33$ stable across three scales.

All code and data are publicly available.

References

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